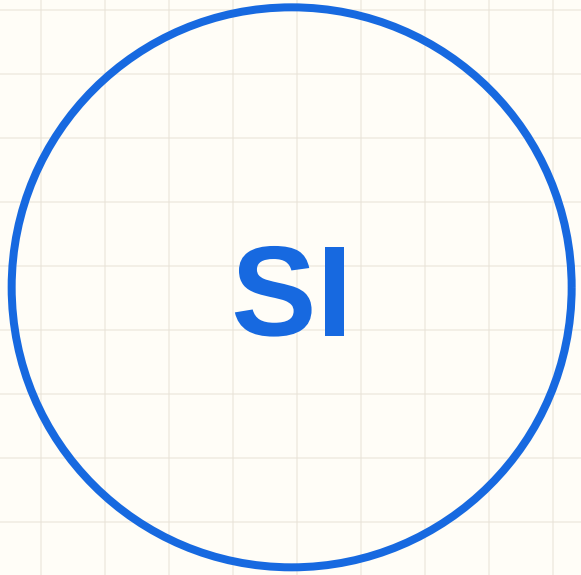
Physical Quantities, SI Units & Unit Conversion



HANDWRITTEN LECTURE

CBSE + NEET

Teacher: _____

Date: _____

OPENING STORY

Why do units matter?

A $125 million unit mistake

TELL THE STORY / DRAW THE SPACECRAFT / COLLECT STUDENT ANSWERS

TOPIC 1

Physical Quantities

Definition:

Examples:

Not physical quantities:

Physical quantity = _____ x _____

TOPIC 2

Fundamental & Derived Quantities

FUNDAMENTAL

Meaning:

Examples:

DERIVED

Meaning:

Examples:

Seven SI Base Quantities

Physical quantity	SI unit	Symbol

TOPIC 4

Derived Quantities & Units

Speed = _____

Acceleration = _____

Force = _____

Work = _____

DERIVE ONE SI UNIT USING BASE UNITS

TOPIC 5

Unit Conversion Method

72 km h^-1 = _____ m s^-1

WRITE THE COMPLETE CONVERSION AND CANCEL UNITS

Shortcut discovered by students: km/h -> m/s = x _____

45-SECOND STUDENT CHALLENGE

90 km h^-1 = ? m s^-1

A 15 m s^-1

B 20 m s^-1

C 25 m s^-1

D 32.4 m s^-1

Student method:

Common Mistakes

1. Number without a unit

Correction: _____

2. Incorrect unit symbol

Correction: _____

3. Wrong conversion direction

Correction: _____

4. Unreasonable final answer

Correction: _____

Practice Board

1. Convert 36 km h^-1 to m s^-1.

2. Write the SI unit of force using base units.

3. A measurement is written as 12.5. What is missing?

Complete Before You Leave

1. **A physical quantity must contain**

2. **The SI base unit of mass is**

3. **To convert km/h to m/s, I**

The Lecture Ends. Learning Continues.

1 **Learn** -> 2 **Attempt** -> 3 **Analyse** -> 4 **Correct** -> 5 **Continue**

EXPLAIN THE DAILY TEST + NEXT-DAY REVIEW CYCLE